

Minimum Lap-Time Optimization of a Racing Line Using Optimal Control

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Problem Formulation

Objective

$$\min_{T, z(\cdot), u(\cdot)} T$$

States and controls

$$z = [s \quad e_y \quad e_\psi \quad v]^T$$

$$u = [a \quad \delta]^T$$

s : track progress,

e_y : lateral offset,

e_ψ : heading error,

v : velocity,

a : longitudinal acceleration,

δ : steering angle.

Vehicle dynamics

$$\dot{s} = \frac{v \cos(e_\psi)}{1 - \kappa e_y},$$

$$\dot{e}_y = v \sin(e_\psi),$$

$$\dot{e}_\psi = \frac{v}{L} \tan(\delta) - \kappa \dot{s},$$

$$\dot{v} = a.$$

Physical and Track Constraints

Track boundaries

$$-e_{y,\max} \leq e_y \leq e_{y,\max}$$

Velocity

$$0 \leq v \leq v_{\max}$$

Longitudinal acceleration

$$a_{\min} \leq a \leq a_{\max}$$

Steering

$$-\delta_{\max} \leq \delta \leq \delta_{\max}$$

Lateral acceleration

$$\left| \frac{v^2}{L} \tan(\delta) \right| \leq a_{y,\max}$$

Frenet validity

$$1 - \kappa e_y > 0$$

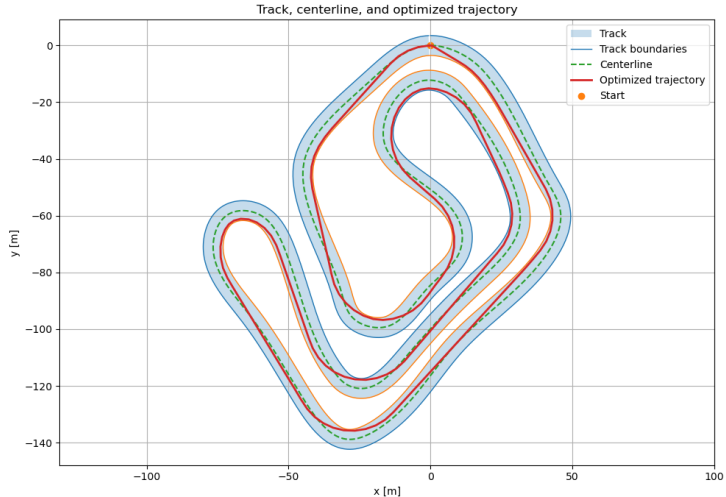
Numerical Solution

Direct multiple shooting

$$Z_{k+1} = F_{\text{RK4}}(Z_k, U_k, \Delta t), \quad \Delta t = \frac{T}{N}$$

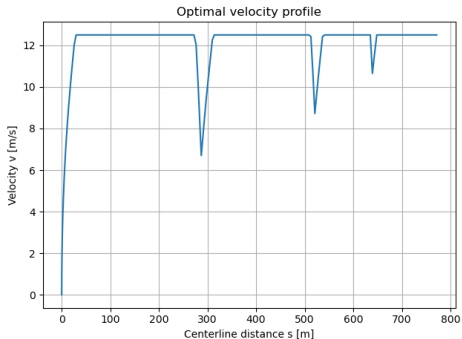
- The lap is divided into $N = 200$ shooting intervals.
- The states Z_k are optimized at every node.
- Vehicle dynamics are integrated with RK4.
- The resulting NLP is built in CasADi and solved with IPOPT.

Optimized Trajectory



The optimized path uses the available track width to increase the effective cornering radius.

Results



Quantity	Value
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Lap time	61.665 s
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Maximum speed	12.5 m/s
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Maximum offset	2.87 m
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IPOPT iterations	357
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Solver time	8.0 s
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The kart reaches the speed limit on most straights and slows down in the high-curvature sections.

Extension: Static Obstacle Avoidance

Effective obstacle radii

$$r_{s,i} = \frac{l_i}{2} + \frac{L}{2} + d_{\text{safe}}, \quad r_{e_y,i} = \frac{w_i}{2} + \frac{b}{2} + d_{\text{safe}}, \quad i = 1, \dots, N_{\text{obs}}$$

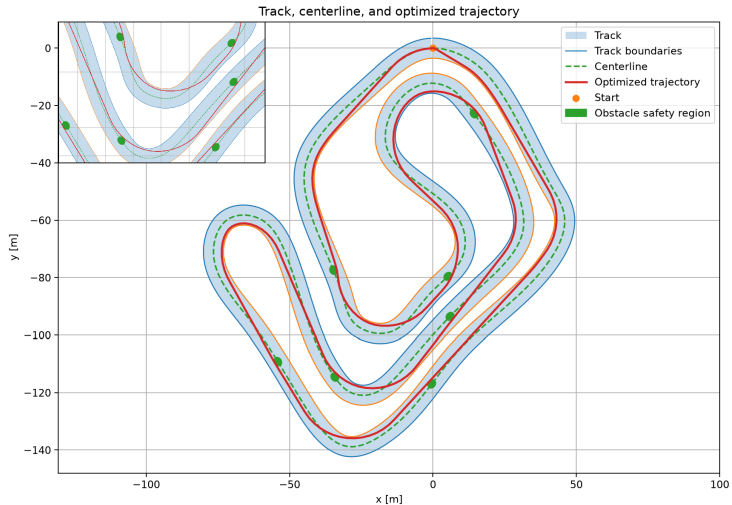
l_i , w_i – obstacle length and width, L , b – vehicle wheelbase and width, d_{safe} – safety margin.

Collision avoidance

$$\left(\frac{s - s_i}{r_{s,i}} \right)^2 + \left(\frac{e_y - e_{y,i}}{r_{e_y,i}} \right)^2 \geq 1, \quad i = 1, \dots, N_{\text{obs}}$$

- Obstacles are represented by elliptical safety regions in Frenet coordinates.
- Constraints are evaluated at shooting nodes and interval midpoints.

Optimized Trajectory with Obstacles



The optimized trajectory deviates only locally to avoid the obstacles, increasing the lap time from 61.665 s to 61.800 s.

Conclusions

- The optimal control problem was successfully solved using direct multiple shooting.
- The optimized trajectory uses almost the full available track width.
- The kart reaches the maximum speed on the straights and slows down in the most demanding corners.
- The resulting minimum lap time is 61.665 s.
- The proposed formulation was successfully extended to include static obstacle avoidance.

Main result

Optimal control provides a feasible racing line that minimizes the lap time while satisfying the vehicle and track constraints.